

Universal BFG Reclosure and the Conditions for Irreducible Emergence

A formal synthesis, an exact novelty theorem, an identifiability result, and a generative test architecture

Marcel Theodor Wende

BalanceFeld Gleichung (BFG) · Independent Researcher

E-mail: marcel-wende@gmx.net | ORCID: 0009-0007-4028-2208

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Terminology. In this manuscript, BFG means BalanceFeld Gleichung. Older source titles retaining “Balance–Field Framework” are cited under their original names.

Scientific status. This is a theoretical preprint. Exact statements are restricted to their hypotheses; exploratory synthetic calculations are not empirical confirmation of a universal natural law.

Abstract

We connect the generator-level BFG grammar of formation, bounded non-vanishing persistence, and relational reclosure to its finite five-object successor representation. A neutral-compatible stratum admits an exact inheritance–novelty identity: commutation of inherited formation with neutral load is equivalent to unchanged formation, while noncommutation produces strict loss of the second spectral moment. This result witnesses internal structural novelty, not automatically an irreducible higher closure. We formalize the latter by source continuity, recursive higher closure, a generative removal intervention, and a prespecified class of mechanistically different rivals. A representation theorem shows that an invertible recoding of a hidden mediator is observationally indistinguishable from the original dynamics; requiring a model to beat its exact clone is therefore impossible. A constructed pilot gives a positive mediator-versus-removal contrast, but a direct dyad performs better and an exact clone ties. We provide a corrected prospective test rule and state the resulting boundary: BFG has a coherent universal closure and emergence criterion, while universal existence and carrier-specific irreducibility remain unproved.

Keywords: BalanceFeld Gleichung; neutral mediation; recursive closure; finite reclosure; structural novelty; causal removal; identifiability; emergence.

1. Introduction and claim discipline

The BFG research line contains two descriptions of closure. The generator-level account states the roles a formation must retain and reapply to relations among closed sources [1]. The finite account constructs a successor from a typed state and may terminate when an exact admissibility gate fails [2]. This paper gives a common logical reading of the two descriptions and separates a generated successor, a structural novelty witness, and an irreducible higher closure. The distinction is needed because the finite theorem concerns the spectrum of an inherited formation operator, whereas the strong emergence claim concerns a new relation among still-active source closures.

The present synthesis introduces no new numerical constant into the BFG finite map. Its removal tests and rival models are criteria for a scientific claim about a particular carrier; they do not constitute an additional BFG generator. Standard linear algebra is used as representation mathematics, not as an external physical premise [11].

Table 1. Claim classes and scope.

Class	What is established	Boundary
Definition	Specifies formation, closure, or emergence predicates.	Does not assert that a carrier satisfies them.
Exact finite theorem	Proves an identity on its stated BFG stratum.	Does not by itself prove a higher source relation.
Synthetic pilot	Compares constructed trajectories and controls.	Does not establish natural realization.
Carrier claim	Requires an independent measurement and intervention.	Not established by the current toy models.

2. Generator-level closure and relational reapplication

Let $D(X)$ express constitutive differentiation, $Coh(X)$ constitutive coherence, $N(X)$ neutral compatibility of their coupling, and $\Gamma(X)$ regulated exclusion of non-integrable excess without erasure of the constitutive roles. We use the source predicates from the pure BFG emergence account [1]:

$$F_BFG(X) := D(X) \wedge Coh(X) \wedge N(X) \wedge \Gamma(X). \quad (1)$$

Persistence requires admissible continuations to preserve a nonzero constitutive signature with bounded responses. Writing this property as $Pers(X)$ gives

$$Cl_BFG(X) := F_BFG(X) \wedge Pers(X). \quad (2)$$

A closed source can become a pole of a relation among several closed sources. The same predicates in (1)–(2) are then applied to the relational candidate rather than assuming in advance that every source pair generates a new level. These equations are grammar-level definitions. They specify what a formation and a recursive closure would be, while a carrier-specific generator must produce the relevant trajectory and relation.

For a source family $S=\{C_1,...,C_m\}$, let R_+ denote a proposed constitutive relation and C_+ the candidate higher closure. $Continuity(S,C_+)$ requires each selected source to remain independently closure-effective in the higher configuration. A one-time correlation or an aggregate readout is insufficient for recursive closure [1,7].

3. The finite reclosure representation

The current finite BFG candidate is represented by five typed objects: a Hermitian formation geometry K , formation density ρ_F , normalized identity witness ρ_W , positive neutral load Y , and persistent projector P [2]. Its successor operation is deterministic on its declared admissible domain and returns an absorbing terminal outcome otherwise:

$$\Xi := (K, \rho_F, \rho_W, Y, P), \quad U_BFG(\Xi) \in \{\Xi_+, \perp\}. \quad (3)$$

The canonical neutral pair is complementary:

$$C_N(Y) = (I+Y)^{-1}, \quad B_N(Y) = Y(I+Y)^{-1}, \quad C_N(Y)+B_N(Y)=I. \quad (4)$$

The selected persistent source, reciprocal branch loads, and actual analysis map A determine the active target carrier. The canonical quadratic neutral completion of the later BFG corpus has the following variational form [3,4]:

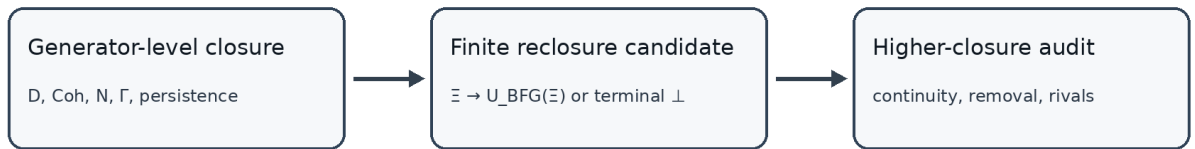
$$J_A(D, \nu) = \frac{1}{2} \|D - A^\dagger \nu\|^2 + \frac{1}{2} \|\nu\|^2. \quad (5)$$

$$\min_\nu J_A(D, \nu) = \frac{1}{2} \langle D, (I + A^\dagger A)^{-1} D \rangle; \quad Y_A = A^\dagger A. \quad (6)$$

Equation (6) follows by solving $(I + AA^\dagger)\nu = AD$ and substituting the minimizer. Y_A is the source-side Gram load. In a declared positive target metric H_+ , the same source-side load grammar reads $Y_A = A^\dagger H_+ A$. On the active carrier, polar identification transports its positive singular spectrum to the target-side successor representation Σ^2 ; the source and target operators are not silently interchanged. The finite successor uses this Gram geometry together with inherited formation transport, identity continuity and a constrained endogenous seed [2,3]. Equation (5) is an explicit canonical quadratic completion; the historical qualitative Level-0 statements alone do not uniquely select its exact form [4,5].

$$\dim \mathcal{H}_+ = \text{rank } A \leq \dim \mathcal{H}. \quad (7)$$

The non-increasing active carrier dimension in (7) refers to this finite map. Structural reorganization within an active carrier is logically different from unbounded creation of new basis directions. The abstract generator can classify relations of closed sources, but mapping a particular physical higher-level carrier into the finite state remains a separate representation obligation.



An admissible successor may be inherited, spectrally novel, or strongly emergent.
Each stronger label requires its own stated assumptions and evidence.

Figure 1. Three distinct operations in the present synthesis. The final audit classifies a generated candidate; it does not add a force to the finite successor map.

4. Exact internal novelty and a counterexample to automatic emergence

On the neutral-compatible persistent stratum, diagonalize the restricted load as $Y_P = \text{diag}(y_i)$. For $\alpha, \beta > 0$, the finite reclosure of formation has a Schur-factor representation [2]:

$$(K_+)_{ij} = \chi_{ij}(K_P)_{ij}, \quad \chi_{ij} = (\alpha + \beta y_i y_j) / \sqrt{[(\alpha + \beta y_i^2)(\alpha + \beta y_j^2)]}. \quad (8)$$

$$1 - \chi_{ij}^2 = \alpha \beta (y_i - y_j)^2 / [(\alpha + \beta y_i^2)(\alpha + \beta y_j^2)]. \quad (9)$$

For Hermitian K_P , the second spectral moment is the sum of squared matrix-entry magnitudes in this basis. Consequently:

$$\text{tr}(K_P^2) - \text{tr}(K_+^2) = \sum_{ij} \alpha \beta (y_i - y_j)^2 |(K_P)_{ij}|^2 / [(\alpha + \beta y_i^2)(\alpha + \beta y_j^2)]. \quad (10)$$

Theorem 1 (inheritance and internal novelty). Under the hypotheses of (8)–(10), $[Y_P, K_P] = 0$ if and only if $K_+ = K_P$. If $[Y_P, K_P] \neq 0$, then $\text{tr}(K_+^2) < \text{tr}(K_P^2)$, so the formation eigenvalue multisets differ.

Proof. Each summand in (10) is nonnegative. It vanishes precisely when $(K_P)_{ij} = 0$ for every pair of distinct neutral-load eigenvalues. Those and only those entries contribute to the commutator $[Y_P, K_P]$. In the zero case, $\chi_{ij} = 1$ on all nonzero entries and $K_+ = K_P$. Otherwise at least one summand is strictly positive. For Hermitian matrices, unequal traces of the square imply unequal spectra. $\square$

The theorem supplies a rigorous internal novelty witness. It neither identifies independent source closures nor proves that a new higher relation is causally necessary. The simple case $Y_P = \lambda I$ gives $\chi_{ij} = 1$ for all i, j and therefore $K_+ = K_P$ in this formation step. Thus even strict internal novelty is not automatic for every admissible step. Repeated application with a frozen neutral geometry tends to attenuate cross-eigenspace terms; the complete BFG orbit must reconstruct its state data if it is to exhibit further novelty [2].

5. A conditional criterion for irreducible higher closure

Fix a source family S , a candidate C_+ , a constitutive relation R_+ , a prespecified decision rule Q , and a prespecified class N^* of mechanistically different rivals. The strong BFG classification proposed in [1], with the causal removal requirement made explicit, is

$$E_BFG(C_+|S; N^*, Q) := \text{Continuity}(S, C_+) \wedge \text{Cl_BFG}(C_+) \wedge \text{Need}(R_+; Q|S) \wedge \text{Excl}(N^*; Q). \quad (11)$$

$\text{Need}(R_+; Q|S)$ requires a removal intervention on the generating relation under matched exogenous conditions that reduces the independently defined higher closure while leaving the sources closure-effective. $\text{Excl}(N^*; Q)$ requires a successful comparison with the defined rival class under the same admissible observations and interventions. The classification is conditional on its null class, decision rule, and intervention set; it is not a theorem that every finite BFG successor satisfies (11).

$$U_BFG(\Xi) = \Xi_+ \neq \perp \not\Rightarrow E_BFG(C_+|S; N^*, Q). \quad (12)$$

A valid successor may be inherited, spectrally novel, or a candidate for irreducible emergence. Strong emergence also requires a separately operationalized relation of still-active sources. Conversely, the abstract generator does not assert that its finite five-object representation is the only possible realization of every carrier.

6. Generative removal and source continuity

For a chosen carrier, write source, mediator, and independent higher-channel dynamics schematically as follows, with u_t denoting measured or controlled exogenous input [8]:

$$s_{t+1} = F(s_t, m_t, u_t; \gamma), \quad m_{t+1} = G(m_t, s_t, u_t; \gamma), \quad h_{t+1} = H(h_t, s_t, m_t, u_t). \quad (13)$$

The removal arm sets the localized coupling γ that forms R_+ to zero in the generator and reruns the entire trajectory from matched initial states with matched exogenous innovations. It must not merely set a reported witness h_t to zero after the fact. Define the prespecified higher recovery score Q so that larger values indicate better closure:

$$\Delta_{\text{gen}} := Q_{\text{full}} - Q_{\text{remove}}. \quad (14)$$

A favorable sign in (14) is interpretable as isolated relation necessity only if source persistence meets its separately prespecified gate in both arms. Reintroducing the same relation at fixed parameters should restore the higher endpoint. A separately measured higher state, rather than a product of source readouts or the removed mediator itself, avoids making the result true by definition. Removal of feedback and export pathways can further distinguish recursive closure from transient coupling, but those are additional targeted interventions [8].

7. Observational equivalence and the clone impossibility result

The distinction between a physical relation and a coordinate description becomes decisive when a hidden mediator is used. Consider a system with mediator m , observed sources s , exogenous inputs u , and observed output y :

$$m_{t+1} = G(m_t, s_t, u_t), \quad y_t = H(m_t, s_t). \quad (15)$$

Let T be an invertible transformation of the hidden mediator state and put $z = T(m)$. The transformed system is

$$z_{t+1} = T[G(T^{-1}(z_t), s_t, u_t)], \quad y_t = H(T^{-1}(z_t), s_t). \quad (16)$$

Theorem 2 (observational clone impossibility). The systems in (15) and (16) have identical observed trajectories for corresponding initial states and the same exogenous inputs. Every statistic based solely on those observed trajectories has the same value.

Proof. Induction on t gives $z_t = T(m_t)$. The output expressions then coincide at each step. Applying any deterministic statistic to equal observed paths gives equal values; the same statement holds for their induced distributions under matched random inputs. Transforming an intervention together with T preserves equivalence. $\square$

A rule demanding strictly better predictive or recovery performance than an exact clone is therefore impossible to satisfy, even for a correctly specified full model. An exact clone tests identifiability of a description and should be reported as a representation-equivalence control, outside the class of rivals that must be beaten. Selective measurement and intervention on a physical mediator, plus explicit exclusions of alternative causal paths, are needed for a claim about its specific physical role. Even then, necessity is relative to the admissible interventions and alternatives.

8. Memory-preserving elimination in the exploratory pilot

The new exploratory synthetic pilot [9] used a mediator with difference $d_t = x_{2,t} - x_{1,t}$, $a = 0.7$ and $b = 0.2$:

$$m_{t+1} = a m_t + b d_t. \quad (17)$$

$$m_t = a^t m_0 + b \sum_{k=0}^{t-1} a^{t-1-k} d_k. \quad (18)$$

Substituting (18) into the source equations constructs a direct coupling of source histories with the same memory and exactly the same source and reporter trajectories. This is a change of representation, not proof that a memoryless direct dyad of comparable complexity is equally effective. It does show why output alone

does not uniquely assign a BFG-specific mediator role. The pilot's memoryless direct dyad is a distinct empirical control and happened to perform better on its recovery endpoint.

The reporter in the pilot had its own update equation and noise, but its input was a constructed function of the two source states. It is an independently evolved synthetic candidate, not an independently established irreducible higher closure. Nor was this pilot a numerical execution of the complete canonical five-object BFG operator.

9. Prespecified rivals and a corrected decision rule

Table 2. Rival classes and what each comparison can establish.

ID	Control	Interpretive role
N1	Common driver without proposed relation	Tests shared-excitation explanations.
N2	Additive source histories	Tests whether separate contributions suffice.
N3	Direct dyadic coupling with comparable memory budget	Tests whether the proposed specific mediation is needed.
N4	Circular shift of one source series	Sensitivity test preserving values and circular autocorrelation.
N5	Matched-capacity latent dynamics	Tests added capacity; exact clones are marked separately.

N1–N3 and mechanistically different instances of N5 are rival explanations. N4 is a temporal sensitivity control, not a generative removal intervention. An exact N5 clone is a representation-control tie by Theorem 2, not a null whose failure can be demanded. Model memory, training data, held-out units, accessible inputs, intervention set, and error tolerance must be frozen before confirmatory data are inspected [8,10].

The decision sequence is: (i) full source continuity and recursive higher recovery; (ii) generative removal lowers the independent higher endpoint with source continuity retained; (iii) restoration recovers the endpoint; (iv) mechanistically different rivals fail to reproduce the same observed and intervention signatures within the prespecified tolerance. For repeated independent units, the prospective protocol specifies a one-sided 95% lower confidence bound for the paired removal effect above zero and Holm correction for the comparisons actually tested [8]. A tie with a true clone is recorded without rejecting a potentially physical mediator.

10. Existing synthetic evidence and its limits

The earlier emergence preprint [1] reported a common-driver toy with source readouts $s_i = \text{clip}(0.25 + 0.7z + \eta_i, 0, 1)$ and a higher witness $h = s_1 s_2$. Its aligned criterion was met in 3,984 of 4,000 runs; a structure-preserving circular-shift control met it in 3,344 of 4,000 (83.6%). The common driver remains an explanation of aligned source variation. Setting the already-computed h to zero does not remove an edge in the source generator.

The later exploratory pilot [9] reran the constructed dynamics after removing its mediator and compared 1,000 paired realizations, each 240 steps long, with a perturbation at step 120. The reporter deviation over 30 post-perturbation steps was smaller when recovery was better. Table 3 reproduces its frozen summary values; all arms passed the pilot's source-spread gate.

Table 3. Exploratory reporter recovery in the constructed pilot; smaller is better.

Model	Mean deviation	Interpretation
Full mediator	0.11793	Baseline constructed mediation.
No mediator / common driver	0.12929	Worse than full by 0.01137.
Direct dyad	0.06058	Better than full by 0.05735.
Exact latent clone	0.11793	Identical paths by construction.

The paired no-mediator minus full contrast was +0.01137 with exploratory bootstrap interval [+0.00895,+0.01358]. The paired direct-dyad minus full contrast was -0.05735 with interval [-0.05868,-0.05608]. On 400 held-out realizations, next-reporter RMSE was 0.015774 for full mediator plus relation and 0.015773 for the direct relational predictor. The exact clone's maximum path difference was zero. These intervals are exploratory resampling descriptions, not prospectively registered confirmation.

The pilot supports only the narrow proposition that the constructed mediator has a causal effect relative to its complete removal in that constructed system. The better direct dyad prevents claiming that the specific mediation is required there. The clone demonstrates a representation tie. Neither result validates universal BFG dynamics in nature.

11. Formal completion and outstanding claims

The strongest presently established synthesis has four parts: a universal recursive grammar (1)–(2); a canonical finite successor candidate (3)–(7); an exact internal novelty theorem on a specified stratum (8)–(10); and a conditional criterion for irreducible higher closure (11) with a workable prospective test architecture. The latter is a complete *definition and decision framework*, not an unconditional theorem that all BFG successors are emergent.

Two independent boundaries remain. First, the canonical quadratic neutral completion is explicit in the later BFG corpus, but it is not uniquely entailed by the earliest qualitative Level-0 statements. A Level-0 formation functional can supply a local positive Hessian and thus a local Gram pullback, while its full quartic expansion need not equal the exact quadratic completion at arbitrary displacement [4,5]. Second, no carrier in the current record supplies a prospectively fixed higher channel, selective relation removal, restoration, source-continuity checks, and successful mechanistically distinct rival comparisons. These are obligations for a chosen realization, not missing algebra in Theorem 1.

$$\text{Universal classification rule} \neq \text{universal existence of irreducible emergence.} \quad (19)$$

A new paper may therefore accurately state that BFG's reclosure and emergence conditions have been integrated formally. It should not state that the integration proves ubiquitous emergence, a unique physical neutral mediator, or empirical universality. A negative or indeterminate carrier result must remain reportable under the same fixed rule.

12. Conclusion

The BFG generator and finite reclosure can be read as complementary layers of one research program. The finite operator constructs candidates, and the generator-level criteria classify their possible higher closure. The neutral-compatible inheritance–novelty identity is exact; the proposed strong emergence criterion additionally requires causal relation necessity and discrimination against genuine rivals. An exact dynamical clone is mathematically indistinguishable from its source representation and belongs in an equivalence class rather than the set of rivals to be defeated. Existing synthetic tests reveal a mediator effect but do not establish the strong claim. Universal formal closure with a precise emergence criterion is achieved; universal emergent realization remains open.

Appendix A. Symbol register

Table 4. Main symbols.

Symbol	Meaning
D, Coh, N, Γ	Constitutive differentiation, coherence, neutral compatibility, regulated exclusion.
$Pers, Cl_BFG$	Non-vanishing bounded persistence and recursive closure.
$\Xi, U_BFG, \perp$	Finite five-object state, successor map, terminal outcome.
K_P, Y_P, χ_{ij}	Restricted inherited formation, neutral load, Schur factor.
S, R_+, C_+	Source closures, proposed constitutive relation, higher closure candidate.
Q, N^*, Δ_{gen}	Prespecified higher endpoint, mechanistic rival class, paired removal contrast.

Appendix B. Reproducibility and provenance

The analytical proofs in Sections 4 and 7 require no simulation. The pilot numbers in Section 10 are transcribed from the frozen report [9] and its accompanying executable Python file `bfg_relational_ablation_pilot.py`, seed 20260929; this paper does not claim a new independent data run. The earlier circular-shift figures are transcribed from [1] and its published numerical supplement. The present Figure 1 is a new explanatory diagram, not data. No natural empirical dataset was analyzed for this manuscript.

Appendix C. Declarations

Author and corresponding author: Marcel Theodor Wende, BalanceFeld Gleichung (BFG), Independent Researcher; marcel-wende@gmx.net; ORCID 0009-0007-4028-2208. Conceptualization, BFG source development, interpretation, and responsibility for the claims: Marcel Theodor Wende. Co-authored precursor works retain their own credited authorship in the references. No human or animal subjects were involved. No external funding or competing-interest assertion is made here; submission-specific disclosures should be confirmed by the author. AI-assisted preparation was used for source consolidation, mathematical cross-checking, English drafting, diagram and document production; scientific responsibility rests with the author.

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